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Software engineer with a research interest in applied LLMs and natural language processing. Recipient of the ICAICTA 2024 Best Paper Award for fine-tuning Llama 2 on sarcasm detection. Currently pioneering AI integration at CSG, shipping LLM-powered systems with prompt engineering, confidence-thresholded inference, and human-in-the-loop safety. Seeking graduate study to pursue rigorous research in AI/ML.

Education

Institut Teknologi Bandung

Bachelor of Engineering in Informatics

August 2020 - October 2024

CGPA: 3.76/4.00 (Cumlaude)

Relevant Coursework: Algorithm and Data Structure, Software Project Management, Database Management, Platform-based Application Development, Parallel and Distributed Systems, Advanced Machine Learning

Publications & Awards

Sarcasm Detection Using Fine-tuned Llama 2

International Conference on Advanced Informatics (ICAICTA 2024)

Singapore

September 2024

Best Paper Award. Fine-tuned **Llama 2** for sarcasm detection on a curated dataset of **21,000+ samples**, **outperforming prior benchmarks by 8%**; explored prompt-engineering and parameter-efficient fine-tuning trade-offs to improve nuanced language understanding.

Work Experience

CSG International

Software Developer Engineer II

Jakarta

April 2026 - Present

CSG is a global provider of customer experience, billing, and payment solutions for the telecom industry.

- Pioneered AI integration at company** by building **AI Email Interpreter** in Java with **LangChain on self-hosted Ollama (qwen3-8b)**, converting raw multilingual client emails into structured order-status-update payloads with **few-shot prompting and confidence-thresholded human-in-the-loop fallback**; delivered the PoC in **3 days for a client hackathon** and **showcased at MWC Barcelona 2026** as the flagship order-to-inbox capability.
- Won 2nd place in internal AI competition** with an **AI-Assisted API Transformation engine** that converts natural-language instructions into transformation scripts, **substantially reducing manual API mapping effort** for customer onboarding.
- Delivered AI Analytics PoC** exposing the production database to business users via an **MCP-backed conversational interface**, enabling self-service exploration without writing SQL.
- Lead developer for Banking & Financial Solutions (BFS)**, driving **10+ international demos including Sibos**, owning the demo environment and acting as **first-line on-call**; shipped supporting AI tooling (a **PDF financial-statement extractor** and an **MCP server for API orchestration**) that accelerated demo turnaround.

Software Developer Engineer I

January 2025 - April 2026

- Earned company TForums DTW 2025 award** by architecting an engine for **intent-driven network API transformations** as MVP of a Catalyst project in **collaboration with Ericsson**.
- Achieved zero transaction failures** by implementing end-to-end recovery with **idempotency, process-engine auto-retry, and enhanced internal error handling**, ensuring **100% transaction reliability** for critical business operations.

Paper.id

Software Engineer

Jakarta

November 2023 - December 2024

Paper.id is a platform providing invoicing, payment, and financing solutions for businesses.

- Accelerated message template creation from hours to under a minute** by integrating third-party SMS and WhatsApp broadcast services with **concurrent Goroutines processing**.
- Architected a unified payment engine** that handles **diverse bank behaviors and merchant-specific requirements** for QRIS, direct transfer, virtual account, and cross-border remittance flows compliant with **Indonesia's Central Bank standards**, achieving a **99% transaction success rate** and contributing **4 billion IDR TPV** in a month.
- Enhanced system robustness** with **dynamic table partitioning and automated transaction reconciliation** over RabbitMQ scheduling and async callbacks, **improving query performance by 20%** and enabling instant reconciliation reports.

Suitmedia

Software Engineer Intern

Bandung

May 2023 - September 2023

- Built a Laravel-based CMS for a client with **35 dynamic menus and 50 API endpoints** from Figma prototypes, redesigning database schemas and ETL flows to **improve query performance by 30%**.

Synpulse

Software Engineer Intern

Jakarta

October 2022 - December 2022

- Supported a **Banking Client's Avaloq core-banking migration** by authoring **20 system specifications** and automating Excel-to-Oracle reconciliation scripts that **cut report processing from 15 minutes to instant**.

Technical Skills

- Programming Languages:** Java, Golang, PHP, Python, JavaScript, Shell Scripting
- Frameworks & Libraries:** Laravel, React, ViteJS, NodeJS, Fastify, JAX-WS
- Databases:** PostgreSQL, MySQL, ArangoDB, Oracle Database
- Tools & Infrastructure:** RabbitMQ, Kafka, OpenTelemetry, Vault, Consul, Docker, Jenkins, Google Cloud Platform, AWS
- AI / ML:** LangChain, Ollama, OpenRouter, Llama 2, Qwen, Model Context Protocol (MCP), Prompt Engineering, LLM Evaluation